PROJECT REPORT

NUMPY ARRAY OPERATIONS

1. Introduction

NumPy is a Python library widely used for numerical computing. It provides arrays and mathematical functions that simplify calculations on numerical data.

This project demonstrates how two NumPy arrays can be used to perform addition, subtraction, multiplication, and division through element-wise operations.

2. Objective

The main objective is to create two NumPy arrays and perform four mathematical operations. The project also aims to understand vectorization and the importance of checking array shapes before performing calculations.

3. Problem Statement

Performing calculations individually for every pair of values can require repetitive code. NumPy provides a convenient way to apply mathematical operations to entire arrays.

This project uses NumPy to perform the required operations and display the results clearly.

4. Tools and Technologies

Python 3 was used to develop the program. NumPy was used to create arrays and perform mathematical operations. Visual Studio Code was used as the development environment, and GitHub was used to store the source code and documentation.

5. Dataset Description

Two numerical arrays were created manually.

Array A: [10, 20, 30, 40, 50]

Array B: [2, 4, 5, 8, 10]

Each array contains five elements. Both arrays have the same shape, allowing corresponding elements to be operated on directly.

6. Methodology

First, the two arrays are created using `numpy.array()`.

Next, the program checks whether both arrays have the same shape. It also validates

the values and prevents division when the divisor array contains zero.

The program performs the following operations:

- Addition: Adds corresponding elements.
- Subtraction: Subtracts each element of Array B from the corresponding element of Array A.
- Multiplication: Multiplies corresponding elements.
- Division: Divides each element of Array A by the corresponding element of Array B.

Finally, the program displays the arrays, their shapes, their data types, and all four results.

7. Results

Operation	Calculated Result
Addition	[12, 24, 35, 48, 60]
Subtraction	[8, 16, 25, 32, 40]
Multiplication	[20, 80, 150, 320, 500]
Division	[5, 5, 6, 5, 5]

Both arrays have the shape (5,).

8. Understanding Vectorization

Vectorization is the process of applying operations to array elements without explicitly writing a loop for each element.

For example, `array_a + array_b` adds corresponding elements of both arrays. NumPy performs the operation across the arrays using its optimized numerical implementation.

Vectorization makes code easier to read and can improve performance for larger numerical datasets.

9. Applications

Array operations are useful in data analysis, scientific calculations, engineering, financial computations, image processing, and machine learning.

For example, similar operations can be used to combine numerical measurements, scale data, and perform calculations on feature arrays.

10. Conclusion

The NumPy Array Operations project successfully demonstrates addition, subtraction, multiplication, and division using two numerical arrays.

The project reinforces the concepts of NumPy arrays, element-wise calculations, array shapes, and vectorization. It provides a foundation for more advanced numerical analysis and machine learning tasks.

11. Project Evidence

Insert a screenshot of the program's actual output from the VS Code terminal.

12. Project Links

GitHub Repository: https://github.com/venky1318/numpy-array-operations